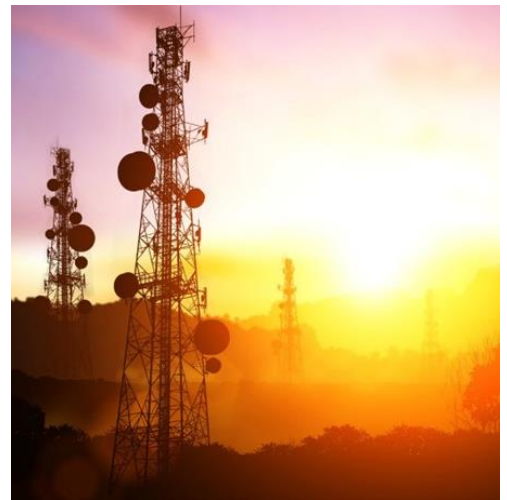


# Model 865B-M

## Wideband Low Noise Synthesizer



### Features

- Low phase noise
- Fast switching down to 20  $\mu$ s
- FM, chirps, pulse
- Internal OCXO, external variable reference

### Applications

- Automated Testing Environments
- LO for frequency converters
- Telecom / SatCom



**Model 865B-M**  
100 kHz to 40 GHz

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## DEFINITIONS

The specifications in the following pages describe the warranted performance of the instrument for  $23 \pm 5$  °C after a 30- minute warm-up period

**Typical:** Expected mean values, not warranted performance

**Min and Max:** Parameter range that is guaranteed by product design, and/or production tested. Warranted performance specifications include guard-bands to account for the expected statistical performance distribution, measurement uncertainties, and changes in performance due to environmental conditions.

## INTRODUCTION

**Ultra-compact, fast and ultra-low phase noise frequency synthesizer with USB & LAN interface.**

The Model 865B-M is a wideband low phase-noise synthesizer settable from 8 kHz to 40 GHz. The settable output power range is from -10 to +25 dBm.

The module has a milli-Hz frequency resolution and uses a high-stability internal reference. The internal reference can be phase-locked to a user-settable external reference. For highest phase coherence, multiple 865B-Ms can be cascaded with just one master reference clock.

The Model 865B-M offers dedicated sweeping capabilities with switching speeds of only 500  $\mu$ s (20  $\mu$ s with option FS) and internal phase and narrow pulse modulation.

The module has USB and LAN interfaces and can be controlled using the SCPI 1999 command set. Operated with an external 24V DC supply, it consumes less than 25 W.

# SPECIFICATIONS

## Signal Specifications

| PARAMETER               | MIN              | TYPICAL                   | MAX              | NOTE                                           |
|-------------------------|------------------|---------------------------|------------------|------------------------------------------------|
| <b>Frequency Range</b>  | 100 kHz<br>8 kHz |                           | 40 GHz<br>40 GHz | Settable from 100 kHz to 43.5 GHz<br>Option 8K |
| Resolution              |                  | 0.001 Hz                  |                  |                                                |
| <b>Phase Resolution</b> |                  | 0.01 deg                  |                  |                                                |
| <b>Switching Speed</b>  |                  | 1.5 ms                    |                  | After SCPI command received                    |
| CW Mode                 |                  | 500 $\mu$ s               |                  |                                                |
| Sweep / List Mode       |                  | 500 $\mu$ s<br>20 $\mu$ s |                  | Option FS                                      |

## Frequency Reference

| PARAMETER                                  | MIN    | TYPICAL               | MAX                           | NOTE                                                        |
|--------------------------------------------|--------|-----------------------|-------------------------------|-------------------------------------------------------------|
| <b>Internal reference frequency</b>        |        | 100 MHz<br>10 MHz     |                               | Option LN                                                   |
| <b>Internal reference output frequency</b> |        |                       |                               |                                                             |
| Temperature stability                      |        |                       | $\pm 100$ ppb<br>$\pm 20$ ppb | 0 to 50 degC<br>Option LN / LN+                             |
| Aging 1 <sup>st</sup> year                 |        |                       | 1 ppm<br>0.03 ppm<br>0.02 ppm | Option LN<br>Option LN+                                     |
| Aging per day                              |        |                       | 5 ppb<br>0.5 ppb<br>< 0.5 ppb | after 30 days operations<br>Option LN<br>Option LN+         |
| Warm-up time                               |        | 5 min                 |                               |                                                             |
| Output of internal reference               |        | 100 MHz               |                               |                                                             |
|                                            |        | 10 / 100 MHz          |                               | Option LN / LN+                                             |
| Output power                               | 0 dBm  | 5 dBm                 |                               |                                                             |
| Output impedance                           |        | 50 Ohms               |                               |                                                             |
| <b>Bypass internal reference input</b>     |        | 100 MHz               |                               | High phase synchronous mode<br>*Options LN/LN+ are bypassed |
| <b>Phase lock to external reference</b>    | 1 MHz  | 10 MHz<br>Integer MHz | 250 MHz                       | Option VREF<br>*Options LN/LN+ are bypassed                 |
| Reference bypass mode                      |        | 100 MHz               |                               |                                                             |
| <b>Reference input level</b>               |        |                       |                               |                                                             |
| 10 MHz or 1-250 MHz                        | -5 dBm | 0 dBm                 | +13 dBm                       |                                                             |
| Bypass 100 MHz                             | 5 dBm  |                       | +15 dBm                       |                                                             |
| <b>Reference input impedance</b>           |        | 50 Ohm                |                               |                                                             |
| <b>Lock Range</b>                          |        |                       |                               |                                                             |
| 10 MHz or 1-250 MHz                        |        |                       | $\pm 1.5$ ppm                 |                                                             |
| Bypass 100 MHz                             |        |                       | >100 ppm                      |                                                             |

## Level Performance

| PARAMETER                 | MIN     | TYPICAL | MAX                | NOTE            |
|---------------------------|---------|---------|--------------------|-----------------|
| <b>Output power level</b> |         |         |                    | (see also plot) |
| 10 MHz to 1.2 GHz         | 0 dBm   |         | +20 dBm<br>+18 dBm | Option FILT     |
| 1.2 GHz to 20 GHz         | -10 dBm |         | +20 dBm<br>+13 dBm | Option FILT     |
| 20 to 30 GHz              | 10 dBm  |         | +20 dBm<br>+13 dBm | Option FILT     |
| 30 to 40 GHz              | 0 dBm   |         | +18 dBm<br>+10 dBm | Option FILT     |
| <b>Power Resolution</b>   |         | 0.5 dB  |                    |                 |
| <b>Power Accuracy</b>     |         | ±0.5 dB | ±2 dB              |                 |

Figure 1: Maximum Output Power (Standard)

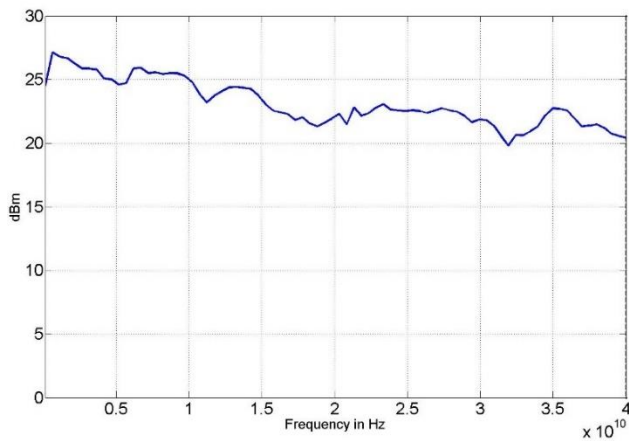
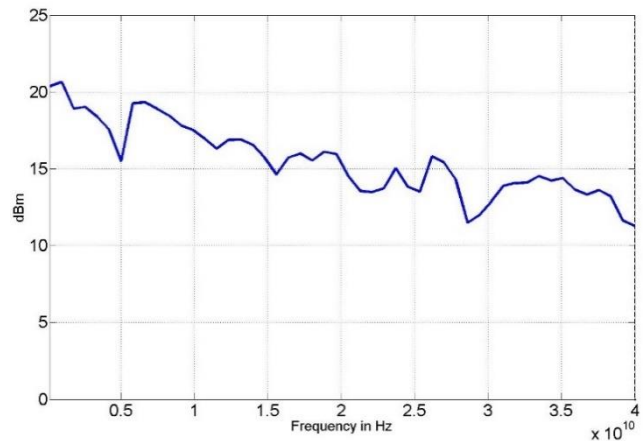


Figure 2: Maximum Output Power with FILT



## Reverse Power Protection and VSWR

| PARAMETER                       | MIN | TYPICAL | MAX     | NOTE |
|---------------------------------|-----|---------|---------|------|
| <b>Reverse power protection</b> |     |         |         |      |
| DC Voltage                      |     | 7 V     |         |      |
| RF Power                        |     |         | +20 dBm |      |
| <b>Output impedance</b>         |     | 50 Ohm  |         |      |
| VSWR                            |     | 1.8     |         |      |

## Phase Noise

| PARAMETER                        | MIN | TYPICAL     | MAX | NOTE            |
|----------------------------------|-----|-------------|-----|-----------------|
| <b>SSB Phase noise at 1 GHz</b>  |     |             |     | (see also plot) |
| At 10 Hz from carrier            |     | -93 dBc/Hz  |     | Option LN / LN+ |
| At 1 kHz from carrier            |     | -130 dBc/Hz |     |                 |
| at 100 kHz from carrier          |     | -145 dBc/Hz |     |                 |
| Wideband noise                   |     | -160 dBc/Hz |     |                 |
| <b>SSB phase noise at 10 GHz</b> |     |             |     |                 |
| At 10 Hz from carrier            |     | -73 dBc/Hz  |     | Option LN / LN+ |
| At 1 kHz from carrier            |     | -110 dBc/Hz |     |                 |
| At 100 kHz from carrer           |     | -125 dBc/Hz |     |                 |

Figure 3: Phase Noise Performance with Option LN

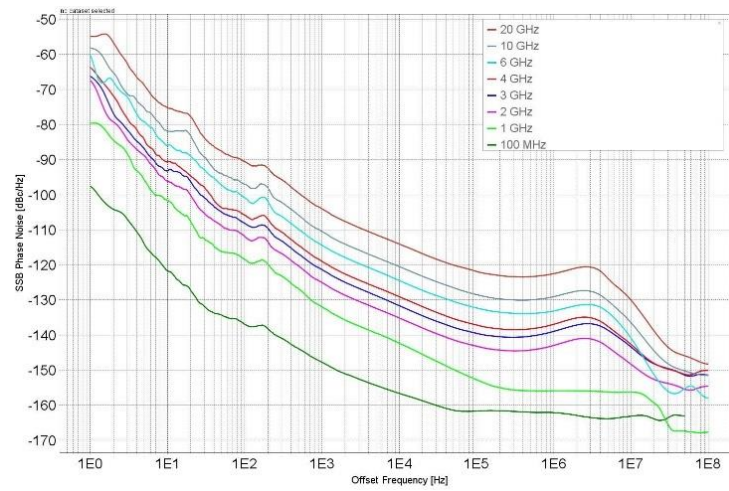


Figure 4: Phase Noise Performance without Option LN (10, 20, 30, 40 GHz)

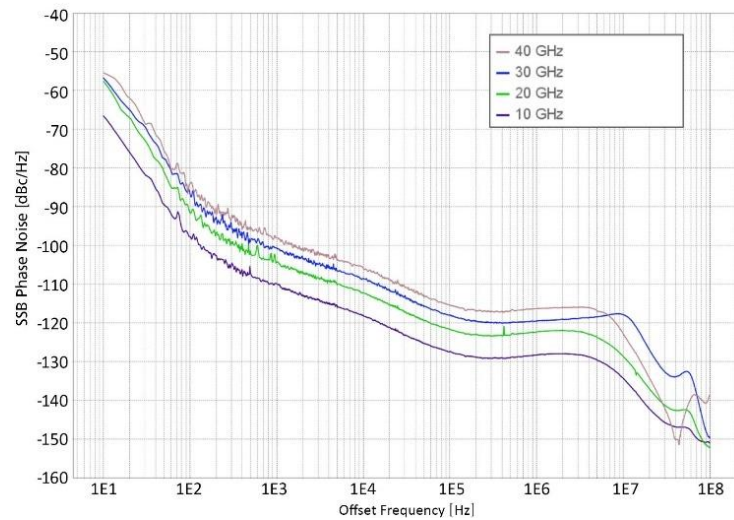
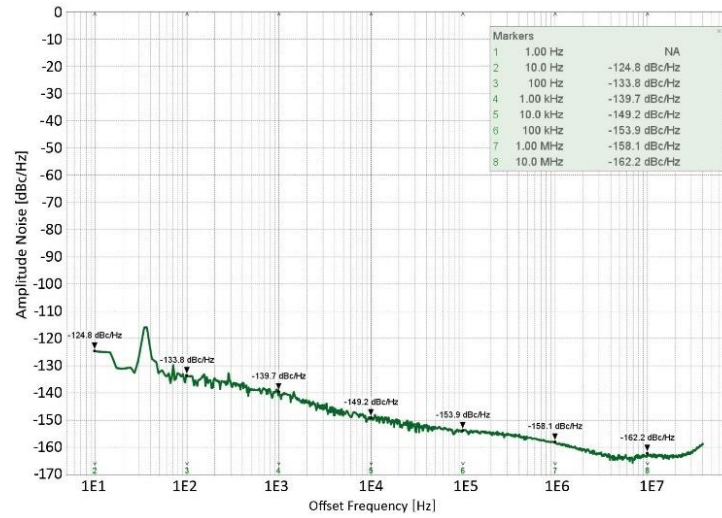


Figure 5: Amplitude Noise at 10 GHz



## Spectral Purity

| PARAMETER             | MIN | TYPICAL            | MAX                | NOTE                 |
|-----------------------|-----|--------------------|--------------------|----------------------|
| Output harmonics      |     | -15 dBc<br>-55 dBc | -45 dBc            | Option FILT, > 1 GHz |
| Sub-harmonics         |     | -75 dBc<br>-50 dBc | -45 dBc<br>-30 dBc | < 20 GHz<br>> 20 GHz |
| Non-harmonic spurious |     | -75 dBc            | -60 dBc            |                      |

Figure 6: Harmonics @ 0 dBm (with option FILT)

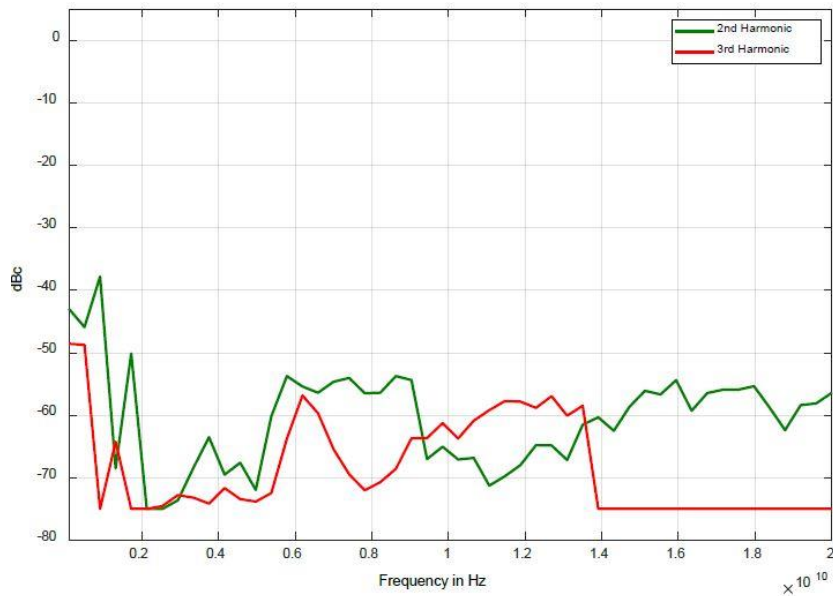
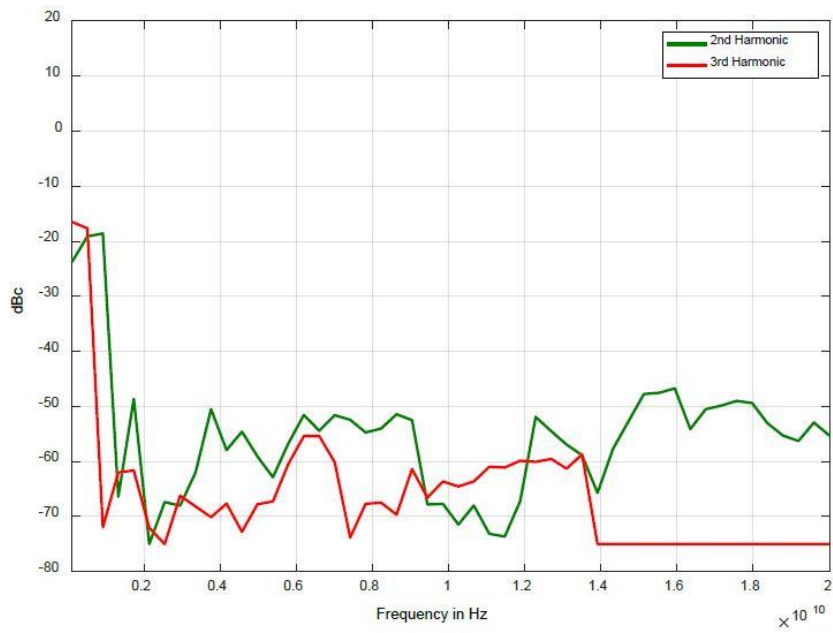


Figure 7: Harmonics +15 dBm (with option FILT)

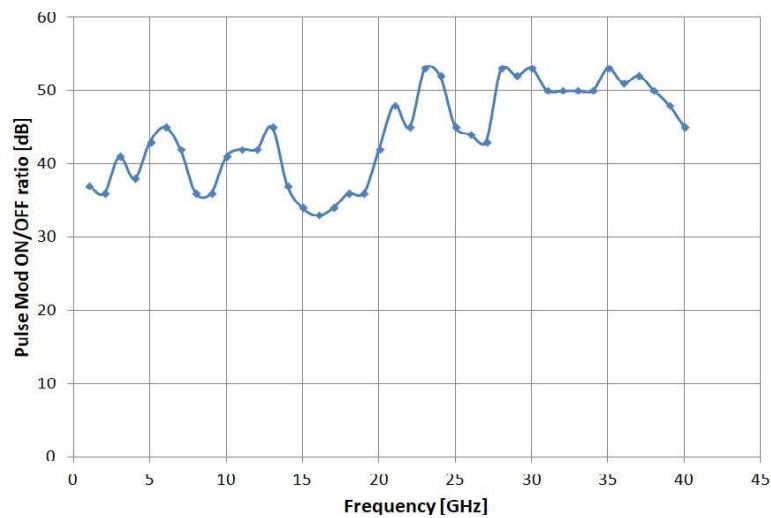


## Modulation Capabilities

### Pulse Modulation

| PARAMETER                    | MIN    | TYPICAL               | MAX    | NOTE                     |
|------------------------------|--------|-----------------------|--------|--------------------------|
| Modulation source            |        | Internal/<br>external |        |                          |
| Pulse rise/fall time         |        | 10 ns                 |        |                          |
| On/off ratio                 |        | 40 dB                 |        | Pout > +10 dBm, see plot |
| Pulse overshoot              |        |                       | 10%    |                          |
| Pulse delay                  |        | 20 ns                 |        |                          |
| Pulse polarity               |        | Normal,<br>inverse    |        | Selectable               |
| External input threshold     | 0.85 V | 0.9 V                 | 0.95 V | TTL compatible           |
| External input voltage range | -0.5 V |                       | +5.5 V | TTL compatible           |
| External input hysteresis    |        | 60 mV                 |        |                          |

Figure 8: Pulse Modulation on-off ratio



### Internal Pulse Generator

| PARAMETER                                   | MIN                        | TYPICAL           | MAX     | NOTE                                 |
|---------------------------------------------|----------------------------|-------------------|---------|--------------------------------------|
| Repetition frequency (PRF)                  | 0.1 Hz                     |                   | 100 MHz | = 1/T                                |
| Duty cycle                                  | 1 % to 99 %<br>in 1% steps |                   |         | within specified minimum pulse width |
| Pulse width settling range                  | 30 ns                      |                   | 5 s     |                                      |
| Pulse pattern modulation &<br>staggered PRF |                            |                   |         | Using internal pattern generator     |
| Programmable pattern length                 | 2                          |                   | 65536   |                                      |
| Duty cycle                                  | 0.05%                      |                   | 99.95%  |                                      |
| Pulse period (T) accuracy                   |                            | 0.00005xT+<br>3ns |         |                                      |
| Pulse width accuracy                        |                            | 0.00005xT+<br>5ns |         |                                      |
| Pulse width resolution                      |                            | 5 ns              |         |                                      |
| Pulse jitter                                |                            | 2 ns              | 5 ns    |                                      |
| Polarity                                    |                            | selectable        |         |                                      |

### Frequency Modulation

| PARAMETER                             | MIN    | TYPICAL     | MAX    | NOTE                                                                                                                                                          |
|---------------------------------------|--------|-------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frequency Modulation</b>           |        |             |        |                                                                                                                                                               |
| Modulation source                     |        | Internal    |        |                                                                                                                                                               |
| Maximum frequency deviation<br>(peak) |        | N · 400 MHz |        | < 1.25 GHz (N=1)<br>1.25 GHz to 2.5 GHz (N=0.125)<br>2.5 GHz to 5 GHz (N=0.25)<br>5 GHz to 10 GHz (N=0.5)<br>10 GHz to 20 GHz (N=1)<br>20 GHz to 40 GHz (N=2) |
| Deviation accuracy                    |        | 0.50%       | 2%     |                                                                                                                                                               |
| Distortion (THD)                      |        | < 1%        |        | 1 kHz rate, 10 kHz deviation                                                                                                                                  |
| Modulation rate                       | 0.1 Hz |             | 80 kHz |                                                                                                                                                               |
| Modulation waveforms                  | Sine   |             |        |                                                                                                                                                               |

### Phase Modulation

| PARAMETER              | MIN    | TYPICAL  | MAX             | NOTE                           |
|------------------------|--------|----------|-----------------|--------------------------------|
| Modulation source      |        | Internal |                 |                                |
| Phase deviation (peak) | 0      |          | 300 · N·<br>rad |                                |
| Deviation accuracy     |        | 0.50%    | 2%              |                                |
| Modulation rate        | 0.1 Hz |          | 80 kHz          |                                |
| Modulation waveforms   |        | Sine     |                 |                                |
| Distortion (THD)       |        | < 1%     |                 | 1 kHz rate & N x rad deviation |



### Sweeping Capability

Sweep type: linear, logarithmic, random

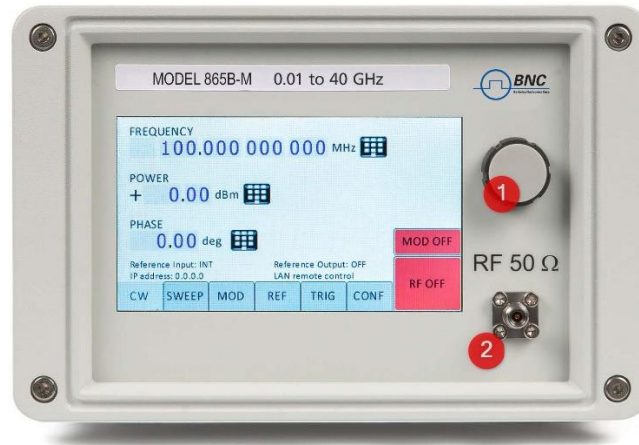
| PARAMETER                        | MIN             | TYPICAL | MAX | NOTE      |
|----------------------------------|-----------------|---------|-----|-----------|
| <b>Frequency sweep</b>           |                 |         |     |           |
| Step time (t <sub>step</sub> )   | 500 μs<br>20 μs |         |     | Option FS |
| Dwell time (t <sub>dwell</sub> ) | 15 μs           |         |     |           |

 **Trigger (TRIG IN): Input is TRIG IN at front panel**

| PARAMETER                                   | MIN                                                 | TYPICAL         | MAX    | NOTE                              |
|---------------------------------------------|-----------------------------------------------------|-----------------|--------|-----------------------------------|
| <b>Trigger types</b>                        | Continuous, single (point), gated, gates direction  |                 |        |                                   |
| <b>Trigger source</b>                       | External, bus (LAN, USB)                            |                 |        |                                   |
| <b>Trigger modes</b>                        | Continuous free run, trigger and run, reset and run |                 |        |                                   |
| Trigger latency                             |                                                     | 5 ns            |        |                                   |
| Trigger uncertainty                         |                                                     | 10 ns           |        |                                   |
| External trigger delay                      | 50 ns                                               |                 | 40 s   |                                   |
| External delay resolution                   |                                                     | 5 ns            |        |                                   |
| <b>Trigger Modulo</b>                       | 1                                                   |                 | 255    | Execute only on Nth trigger event |
| <b>Trigger Polarity</b>                     |                                                     | Rising, falling |        |                                   |
| <b>External trigger input threshold</b>     | 0.85 V                                              | 0.9 V           | 0.95 V | TTL compatible                    |
| <b>External trigger input voltage range</b> | -0.5 V                                              |                 | +5.5 V | TTL compatible                    |
| <b>External trigger input hysteresis</b>    |                                                     | 60 mV           |        |                                   |

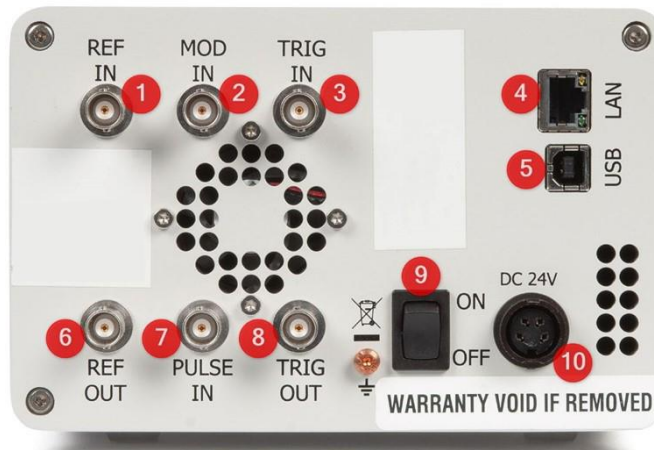
# CONNECTORS

## Front



1. **Rotary Button:** The rotary button is used to change the value selected on the screen.
2. **RF 50Ω connector:** This female K-type connector provides the output for generator signals. The impedance is 50 ohm. The reverse power damage level is +20 dBm maximum. The maximum allowed DC level is +/- 7 V. Please check the data sheet for more details.

## Rear



1. **REF IN** External reference input: BNC female
2. **MOD IN** Modulation input for FM/PM: BNC female
3. **TRIG IN** Trigger input: BNC female
4. **LAN** connection RJ-45
5. **USB 2.0** host and device
6. **REF OUT** Internal reference output: BNC female
7. **PULSE IN** Pulse modulation input: BNC female
8. **TRIG OUT** Trigger output: BNC female
9. **DC power switch**
10. **DC Power plug** (24 V, 3 A)

## ORDERING INFORMATION

| HOST MODEL | PRODUCT      | DESCRIPTION                                                                    |
|------------|--------------|--------------------------------------------------------------------------------|
| 865B-M     | 865B-M       | 40 GHz Synthesizer, desktop enclosure & color touch display                    |
| 865B-M     | Option 8K    | Frequency range extension to 8 kHz                                             |
| 865B-M     | Option LN    | Enhanced close in phase noise & frequency stability                            |
| 865B-M     | Option LN+   | Enhanced close in phase noise & further enhanced long term frequency stability |
| 865B-M     | Option EB    | External power bank adapter cable                                              |
| 865B-M     | Option FILT  | Improved harmonic rejection                                                    |
| 865B-M     | Option IEC   | IEC 17025 calibration with certificate                                         |
| 865B-M     | Option FLASH | MicroSD card slot for removable SD memory                                      |
| 865B-M     | Retrofit     | Applies when options are back-ordered per channel                              |
| 865B-M     | Option VREF  | Variable external reference                                                    |

## GENERAL CHARACTERISTICS

### **Remote programming interfaces**

Ethernet 100BaseT LAN interface, USB 2.0, Control language SCPI Version 1999.0

### **Power requirements 24V $\pm$ 3.0 VDC; 25 W maximum**

**Mains adapter supplied:** 100-240 VAC in/ 24 V 4.0 A DC out

**Environmental** (Levels similar to MIL-PRF-28800F Class 3/4)

**Operating temperature range** 0 to 45 °C

**Storage temperature range** -40 to 70 °C

**Operating and storage altitude** up to 15,000 feet (4600 m)



Safety/EMC complies with applicable Safety and EMC regulations and directives.

**Weight:**  $\leq$  2.2 lbs (1.0 kg) net

**Dimensions (W x L x H):** 6.77 x 11.42 x 4.17 in [172 x 290 x 106 mm]

## DOCUMENT HISTORY

| Version/Status | Date       | Author      | Notes                                                               |
|----------------|------------|-------------|---------------------------------------------------------------------|
| V10            | 2018-05-01 | Jk          | First release                                                       |
| V101           | 2018-08-13 | Jk          |                                                                     |
| V102           | 2018-11-27 | Jk          | Power ranges, pictures, ordering info                               |
| V103           | 2018-12-10 | Mm          | Introduction text                                                   |
| V1031          | 2019-01-28 | Db          | Output of internal reference: 100 MHz                               |
| V104           | 2019-03-18 | Jk          | Corrections                                                         |
| V110           | 2020-09-15 | Jk          | Max power plot                                                      |
| V111           | 2021-02-25 | Db          | Pulse and trigger input electrical specifications                   |
| V112           | 2022-11-18 | Yg          | Add option FILT, TOUCH                                              |
| V113           | 2022-10-02 | Ap/ee       | Added option LN+, updated layout                                    |
| V114           | 2024-02-07 | Ap/mh/yg/at | External reference mode amendment                                   |
| V115           | 2024-08-15 | AT          | New modification the 865B-40T, excluding flange-mount configuration |

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## NOTES